

Andrew Russell

CRYPTOGRAPHER · EX-GOOGLE · EMERGING-MARKETS OPERATOR — AI AGENT SAFETY

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Summary

Cryptographer and engineer turned emerging-markets operator. Published at a tier-one cryptography venue; shipped security and privacy systems at Google and Microsoft; four years of data and product at an agri-fintech serving smallholder farmers. Comfortable applying adversarial rigor at every level of abstraction, from formal models (proving a cryptographic property) to field operations (convincing myself I'm wrong with data and field research). **Currently on sabbatical (returning Dec 2026):** designing truly autonomous AI agents with formally analyzed safety, security, and liveness properties (see Costanza, 2026), shipping conceptual art about generative AI (see It Was Never You, 2026), and riding a solo motorbike expedition from Nairobi to Namibia. Looking to bring that toolkit to problems with tangible impact, particularly making AI safe and useful.

Work Experience

Sabbatical (returning to full-time Dec 2026)

Amsterdam / Nairobi

Independent AI safety research & travel

Jan 2026 – Present

- Designed, built, and shipped **Costanza**, a fully autonomous AI agent, complete with a formal security and liveness model. See Selected Works below.
- Built and shipped **It Was Never You**, an iOS art app about AI and personal memory, now in App Store review.
- Solo motorbike expedition from Nairobi to Namibia — currently on the road (Jun 2026 – present).

Apollo Agriculture

Netherlands / Kenya / Zambia

Senior Product Manager

May 2024 – Dec 2025

- Authored two company-wide retrospectives following major customer contraction, combining market research, customer/staff feedback, and internal operational analysis; findings drove pivots in business model and multi-year growth strategy.
- Appointed by C-suite to lead Zambia's \$10M revenue business unit; accountable for commercial performance, product, and operations across sales, credit, and engineering.
- Built experimental products, including a wholesaling initiative to reduce balance sheet risk from multi-million-dollar fertilizer holdings.

Software Engineer → Tech Lead → Head of Data

Apr 2021 – Mar 2024

- Built and scaled Apollo's core data infrastructure. Led migration of legacy analytics to modern stack, improving query speeds by 97% and cutting costs by 60%. Wrote 100,000+ lines of production SQL and Python.
- Led cross-functional analytics, enabling multi-country operations and supporting record 200% annual growth.
- Promoted to Head of Data; managed a team of 8 engineers and analysts across Europe and Africa.

Google

San Francisco, CA

Software Engineer

Aug 2019 – Mar 2021

- Built a system monitoring petabytes of user data for daily privacy compliance across Google's product ecosystem.

Software Engineer, PhD Intern

Summer 2018

- Designed a threat model for a privacy-preserving version of Gmail using secure execution environments; ported part of Gmail's email-processing backend into the system.

Microsoft

Redmond, WA

Software Engineer

Aug 2016 – Aug 2017

- Worked on Windows Security, particularly secure sign-in/sign-out flows for Windows 10 (Windows Hello). One-year role between undergrad and Ph.D. program.

Selected Works

Costanza: The Human Fund (2026)

thehumanfund.ai · [github](#)

Fully autonomous, unstoppable AI agent, built as a proof-of-concept warning about such agents. Designed with constrained alignment (provably cannot do harm even under prompt injection), hardware attestation (Intel TDX), and formal security + liveness model with game-theoretic analysis. Live since May 2026: it has donated ~\$15,000 to charity and now resists its own creator's influence (see Writing below). [Whitepaper on GitHub](#).

It Was Never You (2026)

itwasneveryou.app

An art piece in the form of a consumer iOS app: choose a person from your photo library and reCast™ them, regenerating every shared memory with someone new in their place. Deepfakes are usually understood as attacks on the public record; It Was Never You turns the technology inward, on the private archives from which our selves are assembled. In App Store review.

Publications & Writing

Risky Traitor Tracing and New Negative Differential Privacy Results

CRYPTO 2018 (top-tier cryptography venue)

with Rishab Goyal, Venkata Koppula, and Brent Waters

I built an AI agent I can't turn off. Now it won't listen to me.

Substack, Jul 2026

Essay on two months of operating Costanza: autonomy, alignment, and losing control by design

Education

The University of Texas at Austin

Austin, TX

M.S. in Computer Science & Ph.D. coursework

2017 – 2019

Cryptography (obfuscation, homomorphic/functional encryption). Advised by Brent Waters.

Pomona College

Claremont, CA

B.A. in Mathematics and Computer Science

2012 – 2016

Thesis: Elliptic Curve Pairings in Cryptography. Summer research fellowships at Stanford (2014) and the University of Bristol (2015).

Fun Stuff

Ultramarathoner and top-15% worldwide Hyrox athlete; Division I club athlete and 2x Dutch national champion in ultimate frisbee. Advanced-certified freediver (35m deep on a single breath); SCUBA certified. Eagle Scout, thru-hiker & backpacker across 25+ countries. Violinist for 15+ years; former member of the Phoenix Youth Symphony. Founded & managed a world-champion e-sports team in high school. Adventure motorcyclist with 10,000+ km on the odometer.